

●MoBiQuant: Problem Setting and Research Gap

Goal: one quantized LLM should adapt precision online as latency and memory budgets change.

Past research: static PTQ

GPTQ, OmniQuant, SmoothQuant, AWQ tune one target bit-width.

They optimize fixed-precision quality, not runtime elasticity.

Changing bit-width usually requires recalibration or quality loss.

Past research: any-precision

AnyPrecisionLLM, AnyBCQ, MatQuant support multiple precisions.

But they add scaling, repacking, table lookups, or kernel overhead.

Switching precision is not yet cheap at token granularity.

What is missing

A single nested model representation.

Token-level precision decisions during inference.

A low-overhead kernel path that fetches only needed bits.

MoBiQuant frames precision as a token-level runtime resource



●Core Observation: Outlier Migration

The tokens that dominate quantization error move when the bit-width changes. Tokens causing high quantization error

Why static calibration fails

Calibration parameters overfit the outlier distribution of one bit-width.

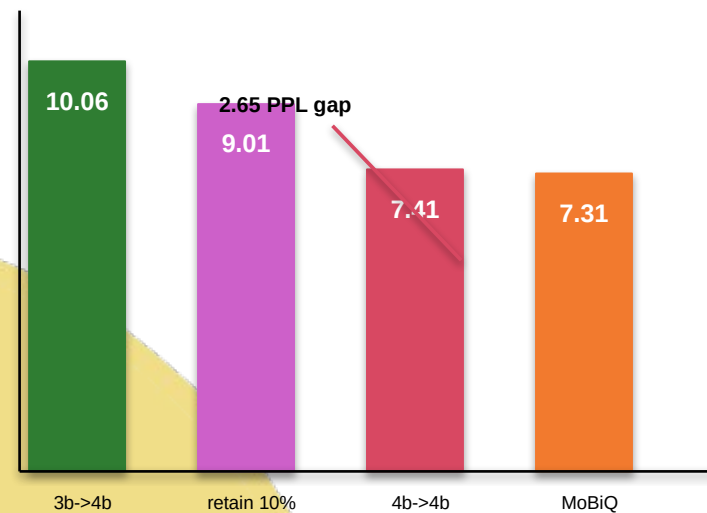
A token well-fitted at 4-bit can become an outlier at 3-bit.

What MoBiQuant exploits

Token sensitivity can guide per-token bit allocation.

Routing high-sensitivity tokens through more bit slices improves generalization.

LLaMA3-8B / WikiText2 PPL

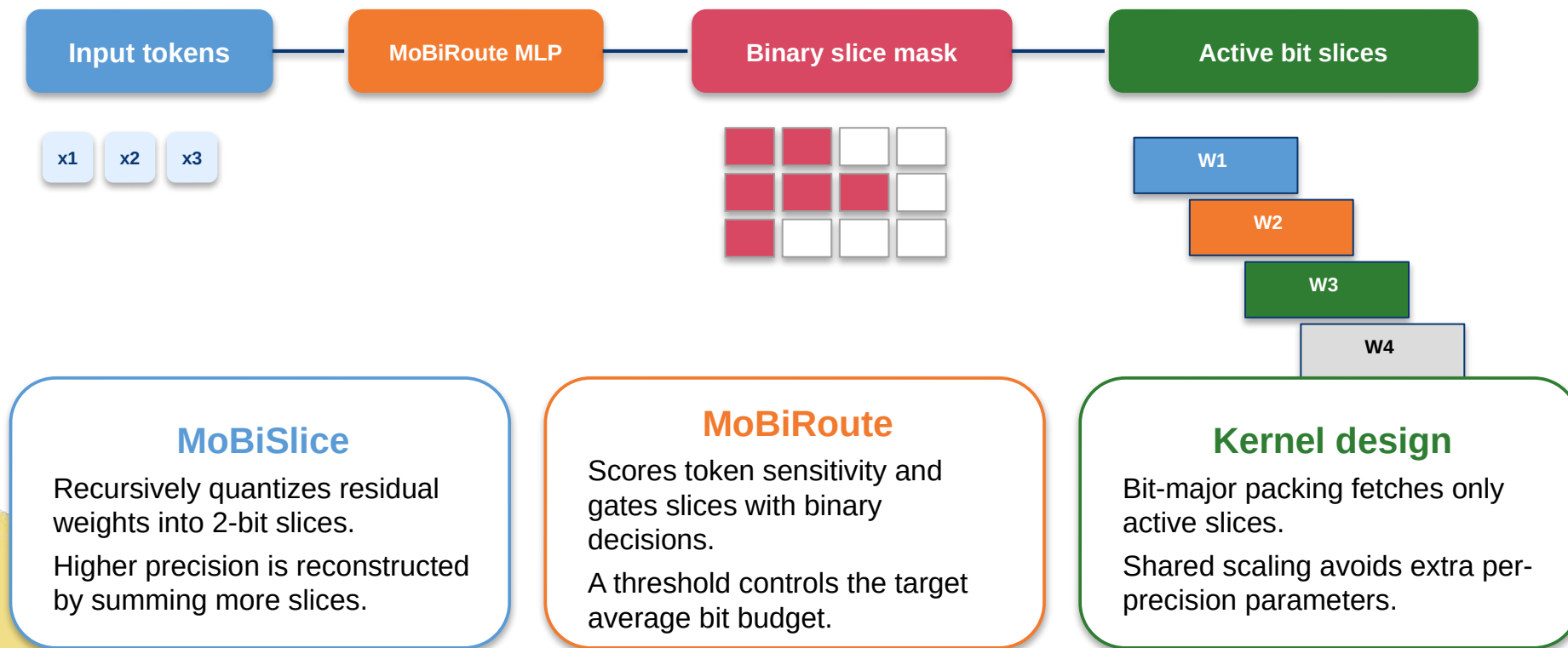


Token-wise error peaks shift across precision



●Method: MoBiSlice and MoBiRoute

A single linear block becomes a routed mixture of residual bit slices.



●Main Results and Takeaways

MoBiQuant keeps elasticity while approaching fixed-precision PTQ quality.

Accuracy

Matches or surpasses static scalar PTQ across LLaMA2 and LLaMA3 model families.
At 2-3 bit, reduces severe PPL collapse versus prior any-precision baselines.

Throughput

Average decoding speedup: 33.8% over AnyPrecisionLLM.
Average decoding speedup: 22.8% over AnyBCQ.

Memory

One nested model replaces multiple per-bit deployments.
Reported memory footprint reduction is up to 3.5x.

Conclusion

The core win is not just storing more precisions.
The router mitigates outlier migration by assigning precision per token.

Takeaway: precision becomes a dynamic information budget at inference time

